

# Finite-Time Feedback Control with Kinetic Compliance

Control, information, and irreversibility with a single measurable affinity

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## Abstract

We develop a finite-state reference theory for feedback control of a binary population using one measurable directional affinity  $h$  and one kinetic compliance parameter  $\gamma$ . The derivation is deliberately finite-time and does not assume equilibrium or stationarity. Starting from the population ensemble  $p_k(n)$ , a hypergeometric sensor and stochastic feedback policy determine the advocacy probability. A reversible finite-compliance controlled kernel then yields an exact susceptibility, an exact state-local action-to-population information  $T_\pi(n)$ , and a closed mean controlled current. Its forward/reverse transition ratio produces a pathwise identity in which the finite-time system entropy change, the affinity-weighted current, and the sensing information sum exactly to the Kullback–Leibler irreversibility of the feedback path measure. Separating the transient system-entropy storage term from this identity leaves the non-storage finite-time control expenditure  $C_{\text{th}} = hJ_c + I(n_k; Y_k)$  and therefore the bounded thermodynamic control efficiency  $\eta_{\text{th}} = hJ_c/[hJ_c + I(n_k; Y_k)] \leq 1$  for target-directed operation. The report also preserves the state-local information analysis, an operational calibration from microscopic controlled LLM transitions, and the current–irreversibility view as a complementary diagnostic.

The conceptual chain used throughout is

$$\boxed{\text{sensing/policy} \longrightarrow \chi_{h,\gamma} \longrightarrow T_\pi(n) \longrightarrow J_c \longrightarrow \Sigma \longrightarrow \eta_{\text{th}}.} \quad (1)$$

Thermodynamic logarithms are natural logarithms, so entropy and irreversibility in the second-law identity are measured in nats. The state-local information  $T_\pi(n)$  is reported in bits, matching the information-theoretic analysis convention used for the empirical action channel.

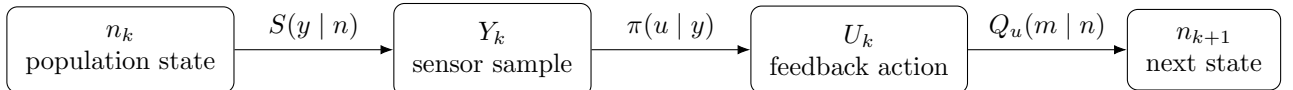
## 1 Problem and one-cycle architecture

Consider a population of  $N$  binary agents and fix a controller target  $Z$ . At feedback cycle  $k$ , let

$$\boxed{n_k \in \{0, 1, \dots, N\}, \quad x_k = \frac{n_k}{N},} \quad (2)$$

where  $n_k$  is the number of agents supporting  $Z$  immediately before sensing and control.

A single feedback cycle has the causal order



Equivalently,

$$\boxed{n_k \longrightarrow Y_k \xrightarrow{\pi} U_k \longrightarrow n_{k+1}.} \quad (3)$$

The controller policy is the conditional action law

$$\boxed{\pi(u \mid y) = P(U_k = u \mid Y_k = y).} \quad (4)$$

The variables and parameters used below are:

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|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| $N$                  | population size                                                                               |
| $n_k, x_k = n_k/N$   | target count and target fraction before cycle $k$                                             |
| $q_c$                | sensor sample size: number of agents read by the controller                                   |
| $Y_k$                | number of target supporters in the sensor sample                                              |
| $U_k \in \{0, 1\}$   | feedback action: 0 = NoOp, 1 = advocate the target                                            |
| $\theta$             | feedback threshold: location where the policy tends to switch                                 |
| $\beta$              | policy gain: sharpness of that switch                                                         |
| $b$                  | number of controlled microscopic opportunities when the controller acts                       |
| $h$                  | directional affinity: asymmetry favoring target-directed versus reverse controlled revisions  |
| $\gamma \in [0, 1]$  | kinetic compliance: how readily a controlled opportunity produces any revision                |
| $\chi_{h,\gamma}(x)$ | controller-induced susceptibility of the mean target fraction                                 |
| $T_\pi(n)$           | state-local action-to-population information                                                  |
| $J_{c,k}$            | mean controlled target-count current in cycle $k$                                             |
| $\Sigma_k$           | finite-time path irreversibility for cycle $k$                                                |
| $C_{\text{th},k}$    | non-storage finite-time control expenditure, $\Sigma_k - \Delta S_{\text{sys},k}$             |
| $\eta_{\text{th},k}$ | thermodynamic control efficiency: directed current divided by non-storage control expenditure |

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**Scope.** The theory is a coarse-grained reference model for the controlled population coordinate  $n_k$ . It does not assign an energetic thermodynamics to the complete language-model population. The role of  $h$  is operational: it is the log directional odds of the controlled revision channel after the combinatorial population multiplicity is separated out. The role of  $\gamma$  is kinetic. This distinction is the central simplification of the model.

## 2 Population ensembles and finite-time notation

The population state is random across repeated realizations. If a derivation or theoretical figure begins from an initial ensemble, we write

$$\boxed{p_0(n) = P(n_0 = n)} \quad (5)$$

and call  $p_0$  the *initial population-state distribution*.

At a generic cycle  $k$ ,

$$\boxed{p_k(n) = P(n_k = n)} \quad (6)$$

is the ensemble distribution before sensing and control, while

$$\boxed{p_{k+1}(m) = P(n_{k+1} = m)} \quad (7)$$

is the ensemble distribution after that feedback cycle. These distributions are not extra model parameters. They summarize the frequencies of population states across repeated realizations at the corresponding cycle.

For readability, a local one-cycle derivation may suppress time indices. Whenever that is done below,

$$\boxed{p(n) \equiv p_k(n), \quad p'(m) \equiv p_{k+1}(m).} \quad (8)$$

No equilibrium or stationarity assumption is made. In general,

$$\boxed{p_{k+1} \neq p_k.} \quad (9)$$

The resulting finite-time change of the population ensemble is retained explicitly in the system entropy change derived later.

For the illustrative current-versus-irreversibility plot we will choose

$$\boxed{p_0(n) = \text{Binomial}(N, 0.25),} \quad (10)$$

only to provide a concrete theoretical ensemble. None of the identities below requires this particular choice.

### 3 Sensing and feedback decision

The controller samples  $q_c$  agents without replacement. If  $Y_k$  is the number of target supporters in the sample, then conditioned on  $n_k = n$ ,

$$Y_k \mid n_k = n \sim \text{Hypergeom}(N, n, q_c), \quad (11)$$

with sensor kernel

$$S(y \mid n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (12)$$

After observing  $Y_k = y$ , the controller advocates with probability

$$\boxed{\pi(1 \mid y) = \sigma \left[ \beta \left( \theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}.} \quad (13)$$

Thus  $\theta$  sets the sensed target-share threshold around which the action changes, while  $\beta$  controls how sharply it changes.

The advocacy probability conditional on the true population state is obtained by averaging the policy over the sensor noise:

$$\boxed{a_n \equiv P(U_k = 1 \mid n_k = n) = \sum_y S(y \mid n) \pi(1 \mid y).} \quad (14)$$

This finite sum is where the sensing resource  $q_c$  and policy parameters  $(\theta, \beta)$  enter the state-dependent action probability.

### 4 Finite-bias, finite-compliance microscopic actuation

Use one directional affinity  $h$  and define

$$p_h = \sigma(h). \quad (15)$$

At one controlled microscopic opportunity, a focal agent is sampled uniformly. If the focal agent is currently non-target, it switches to the target with probability  $\gamma p_h$ . If it is currently target, it switches away with probability  $\gamma(1 - p_h)$ . Therefore

$$K_{h,\gamma}(n+1 | n) = \gamma \frac{N-n}{N} p_h, \quad (16)$$

$$K_{h,\gamma}(n-1 | n) = \gamma \frac{n}{N} (1 - p_h), \quad (17)$$

$$K_{h,\gamma}(n | n) = 1 - K_{h,\gamma}(n+1 | n) - K_{h,\gamma}(n-1 | n). \quad (18)$$

The two parameters have deliberately different jobs:

$$\boxed{h = \text{directional asymmetry}, \quad \gamma = \text{kinetic compliance}.} \quad (19)$$

The affinity  $h$  controls which direction is preferred when a controlled revision occurs. The compliance  $\gamma$  controls how readily a controlled opportunity produces a revision at all.

## 5 Exact susceptibility: derivation of $\chi_{h,\gamma}(x)$

We now derive the exact mean response produced by  $b$  controlled opportunities. The result is

$$\boxed{\chi_{h,\gamma}(x) = [\sigma(h) - x] \Lambda_{b,\gamma}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b.} \quad (20)$$

### 5.1 One controlled opportunity

Suppose the target fraction immediately before the  $j$ th controlled opportunity is  $x_j$ . The count changes by +1 with probability

$$P(\Delta n = +1 | x_j) = \gamma(1 - x_j)p_h, \quad (21)$$

and by -1 with probability

$$P(\Delta n = -1 | x_j) = \gamma x_j(1 - p_h). \quad (22)$$

Consequently,

$$\mathbb{E}[\Delta n | x_j] = \gamma(1 - x_j)p_h - \gamma x_j(1 - p_h) \quad (23)$$

$$= \gamma[p_h - x_j]. \quad (24)$$

This is already useful: the expected microscopic current points toward  $p_h = \sigma(h)$ .

Since  $x = n/N$ ,

$$\mathbb{E}[\Delta x | x_j] = \frac{\gamma}{N}[p_h - x_j]. \quad (25)$$

Hence

$$\mathbb{E}[x_{j+1} | x_j] = x_j + \frac{\gamma}{N}(p_h - x_j) \quad (26)$$

$$= \left(1 - \frac{\gamma}{N}\right) x_j + \frac{\gamma}{N} p_h. \quad (27)$$

Define

$$\alpha_\gamma = 1 - \frac{\gamma}{N}. \quad (28)$$

Then

$$\mathbb{E}[x_{j+1} | x_j] = \alpha_\gamma x_j + (1 - \alpha_\gamma) p_h. \quad (29)$$

## 5.2 Iterating exactly $b$ controlled opportunities

Let  $m_j = \mathbb{E}[x_j \mid x_0 = x]$ . Taking expectations of the affine recursion gives

$$m_{j+1} = \alpha_\gamma m_j + (1 - \alpha_\gamma) p_h. \quad (30)$$

Subtract the fixed point  $p_h$ :

$$m_{j+1} - p_h = \alpha_\gamma (m_j - p_h). \quad (31)$$

Iterating  $b$  times,

$$m_b - p_h = \alpha_\gamma^b (x - p_h), \quad (32)$$

so

$$m_b = p_h + (x - p_h) \alpha_\gamma^b \quad (33)$$

$$= x + (p_h - x)(1 - \alpha_\gamma^b). \quad (34)$$

Therefore

$$\mathbb{E}[x_b - x \mid x] = (p_h - x) \left[ 1 - \left( 1 - \frac{\gamma}{N} \right)^b \right]. \quad (35)$$

If ordinary  $q = 1$  voter updates are interspersed between controlled opportunities, their conditional mean obeys

$$\mathbb{E}[x_{j+1} \mid x_j, \text{ordinary}] = x_j. \quad (36)$$

They are therefore mean-neutral and do not change this exact advocacy-versus-NoOp mean response. We define

$$\boxed{\chi_{h,\gamma}(x) \equiv \mathbb{E}[x' \mid U = 1, x] - \mathbb{E}[x' \mid U = 0, x] = [\sigma(h) - x] \Lambda_{b,\gamma}.} \quad (37)$$

The directional affinity fixes the preferred set point, while kinetic compliance controls the response amplitude. The zero-response point is

$$\boxed{x^* = \sigma(h).} \quad (38)$$

## 6 Exact state-local action-to-population information

The susceptibility measures a signed mean response. A complementary question is whether the controller action is statistically visible in the next population state. This is captured by the exact state-local action-to-population information.

### 6.1 Definition

For a fixed pre-control state  $n = n_k$ , use the shorter one-cycle notation

$$m \equiv n_{k+1}. \quad (39)$$

Define the two controlled-layer next-state kernels

$$Q_0(m \mid n) = \delta_{mn}, \quad Q_1(m \mid n) = K_{h,\gamma}^b(m \mid n). \quad (40)$$

With  $a_n = P(U_k = 1 \mid n_k = n)$ , the action-marginalized kernel is

$$Q_\pi(m \mid n) = (1 - a_n) Q_0(m \mid n) + a_n Q_1(m \mid n). \quad (41)$$

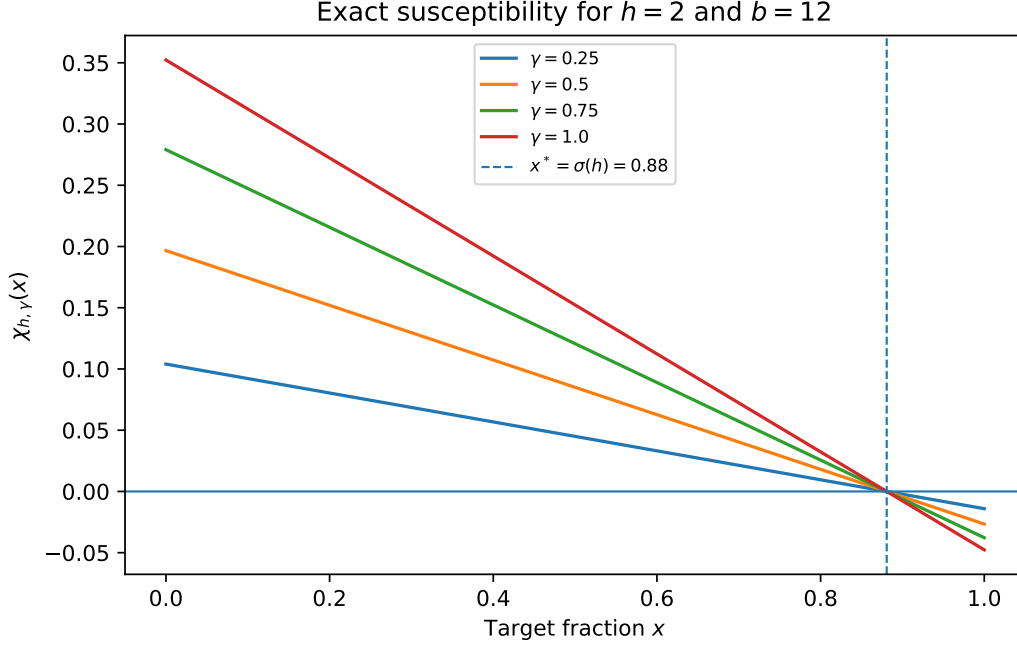


Figure 1: Exact susceptibility for  $h = 2$  and  $b = 12$ . Finite kinetic compliance changes the response amplitude through  $\Lambda_{b,\gamma}$  without moving the zero-response point  $x^* = \sigma(h)$ .

The state-local action-to-population information is

$$T_\pi(n) = I(U_k; n_{k+1} \mid n_k = n) = \sum_{u \in \{0,1\}} P(u \mid n) \sum_m Q_u(m \mid n) \log_2 \frac{Q_u(m \mid n)}{Q_\pi(m \mid n)}. \quad (42)$$

Thus, in the local notation  $m = n_{k+1}$ , this is exactly  $I(U; m \mid n)$ .

The subscript  $\pi$  emphasizes that the information channel is induced by the feedback policy. The quantity is large only when two conditions hold simultaneously:

1. the action remains variable at that state, so  $a_n$  is not too close to 0 or 1;
2. the ADVOCATE and NoOp next-state kernels are sufficiently separated.

If either condition fails, observing the next population state reveals little about whether the controller acted.

If an ensemble-weighted quantity is desired at cycle  $k$ , then

$$I(U_k; n_{k+1} \mid n_k) = \sum_n p_k(n) T_\pi(n). \quad (43)$$

The summary figure below instead uses the simple arithmetic state average  $(N+1)^{-1} \sum_n T_\pi(n)$  so that it describes the channel geometry without choosing an occupancy distribution.

**Do not confuse the two information quantities.** The sensing information introduced later in the finite-time second law is

$$I(n_k; Y_k) = \text{sensing information}, \quad (44)$$

whereas

$$T_\pi(n) = I(U_k; n_{k+1} \mid n_k = n) = \text{action-to-population information}. \quad (45)$$

The first quantifies what the sensor learns about the current population; the second quantifies how visible the action is in the next population state.

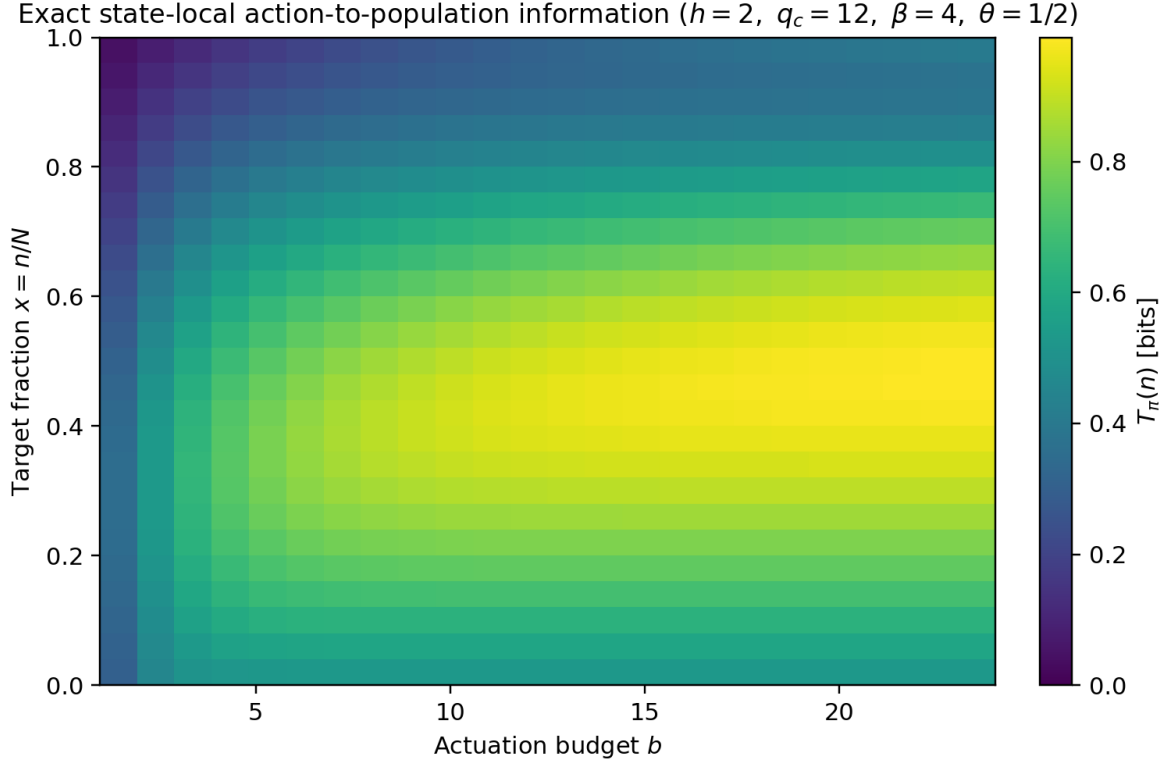


Figure 2: Exact state-local action-to-population information for  $N = 24$ ,  $h = 2$ ,  $q_c = 12$ ,  $\beta = 4$ ,  $\theta = 1/2$ , and  $\gamma = 1$ . Increasing  $b$  separates the advocacy and NoOp kernels more strongly, but information remains small in states where the policy action is nearly deterministic.

## 6.2 A bounded information–response conversion

The state-local information also provides a rigorous ceiling on how much coherent mean motion can be associated with the action-dependent next-state distributions. At fixed  $n$ ,  $T_\pi(n)$  is the weighted Jensen–Shannon divergence between  $Q_0(\cdot | n)$  and  $Q_1(\cdot | n)$  with mixing weight  $a_n$ . Pinsker’s inequality, followed by the fact that the target fraction  $m/N$  lies in  $[0, 1]$ , gives

$$T_\pi(n) \geq \frac{2a_n(1 - a_n)}{\ln 2} \chi_{h,\gamma}(n/N)^2. \quad (46)$$

Equivalently, whenever  $T_\pi(n) > 0$ , the bounded information–response efficiency is

$$\eta_{\text{IR}}(n) \equiv \frac{2a_n(1 - a_n)\chi_{h,\gamma}(n/N)^2}{(\ln 2)T_\pi(n)} \leq 1. \quad (47)$$

This ratio has a different role from the thermodynamic efficiency derived later. It asks whether action-dependent distributional change is expressed as coherent mean target motion; it does not account for the sensing information acquired by the controller or for path irreversibility.

## 6.3 Baseline landscape

Figure 2 shows the exact local landscape as a function of the current target fraction  $x = n/N$  and actuation budget  $b$ , for the baseline values  $N = 24$ ,  $h = 2$ ,  $q_c = 12$ ,  $\beta = 4$ ,  $\theta = 1/2$ , and  $\gamma = 1$ .

## 6.4 Parameter sweeps

Figures 3–6 vary  $h$ ,  $q_c$ ,  $\beta$ , and  $\theta$  one at a time while holding the other parameters at their baseline values.

## 6.5 How to read the parameter dependence

The same interpretation as in the state-local picture is useful throughout. Increasing  $h$  primarily changes *kernel separation*. Changing  $q_c$ ,  $\beta$ , or  $\theta$  primarily changes *action variability through sensing and policy*. Large  $T_\pi(n)$  requires both. This is why a stronger controller does not produce a uniformly larger information channel at every state: if a policy becomes deterministic, its action entropy can collapse even when its action has a strong behavioral effect.

## 7 Controlled current

We now convert the susceptibility into a target-directed controlled current. Define

$$Q_1 = K_{h,\gamma}^b, \quad Q_0 = I. \quad (48)$$

For one cycle, the controlled current increment is the signed target-count displacement when the controller acts,

$$j_c = U_k(n_{k+1} - n_k). \quad (49)$$

Conditioned on state  $n$  and advocacy,

$$\mathbb{E}[j_c \mid n_k = n, U_k = 1] = N\chi_{h,\gamma}(n/N) \quad (50)$$

$$= N\Lambda_{b,\gamma} \left[ \sigma(h) - \frac{n}{N} \right]. \quad (51)$$

The action is only taken with probability  $a_n$ , and the pre-sensing state is distributed as  $p_k(n)$ . Averaging therefore gives

$$J_{c,k} = N\Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[ \sigma(h) - \frac{n}{N} \right]. \quad (52)$$

When the cycle index is suppressed we write  $J_c \equiv J_{c,k}$ . Positive  $J_c$  means that controlled revisions produce a net movement toward the controller target on average; negative  $J_c$  means the net controlled movement is away from it.

## 8 Transition-ratio identity

The key algebraic property is that  $\gamma$  changes the kinetics but cancels from the directional forward/reverse ratio. For neighboring count states,

$$\frac{K_{h,\gamma}(n+1 \mid n)}{K_{h,\gamma}(n \mid n+1)} = \frac{\gamma^{\frac{N-n}{N}} p_h}{\gamma^{\frac{n+1}{N}} (1-p_h)} \quad (53)$$

$$= \frac{N-n}{n+1} \frac{p_h}{1-p_h} \quad (54)$$

$$= \frac{N-n}{n+1} e^h. \quad (55)$$



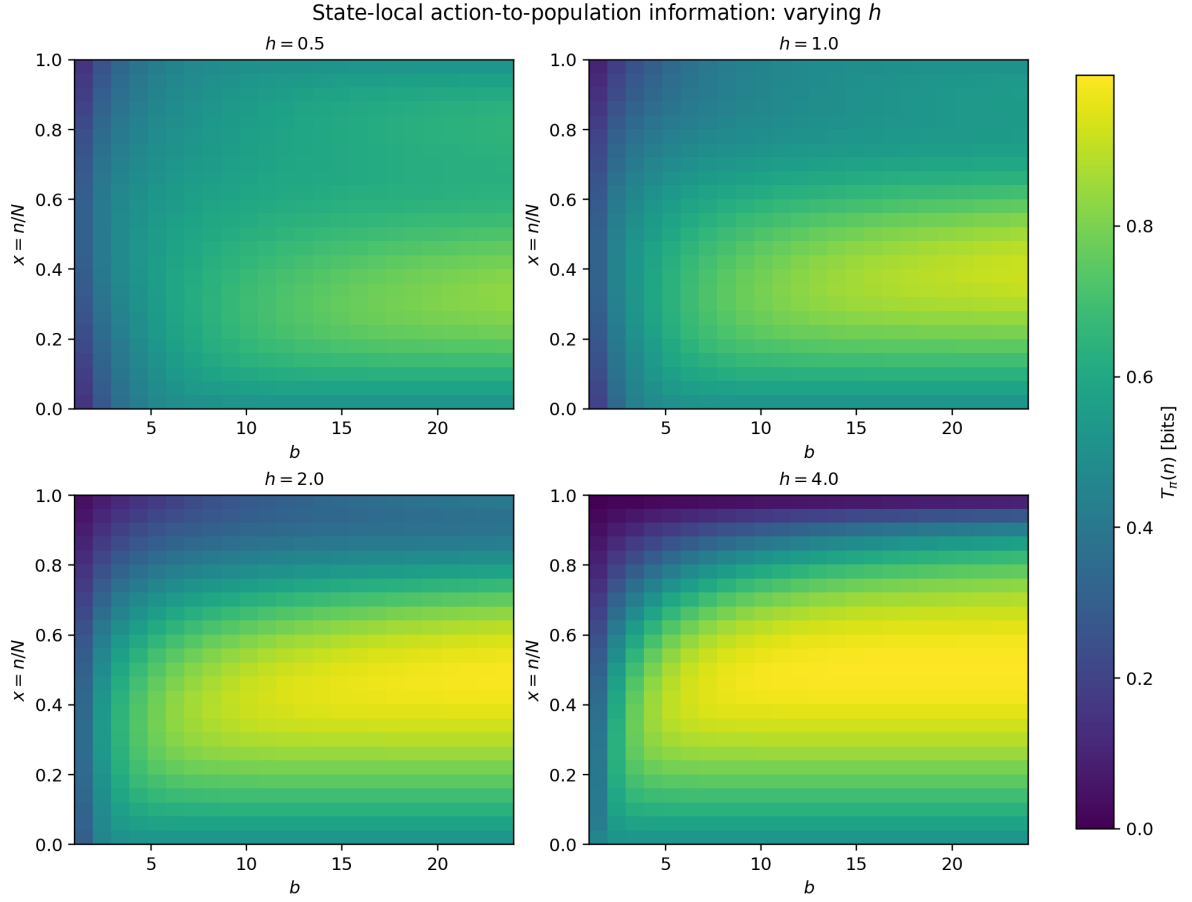


Figure 3: Exact state-local action-to-population information while varying the directional affinity  $h$ . Larger  $h$  changes and generally strengthens separation of the advocacy kernel from NoOp, with the enhancement concentrated where the policy retains nontrivial action entropy.

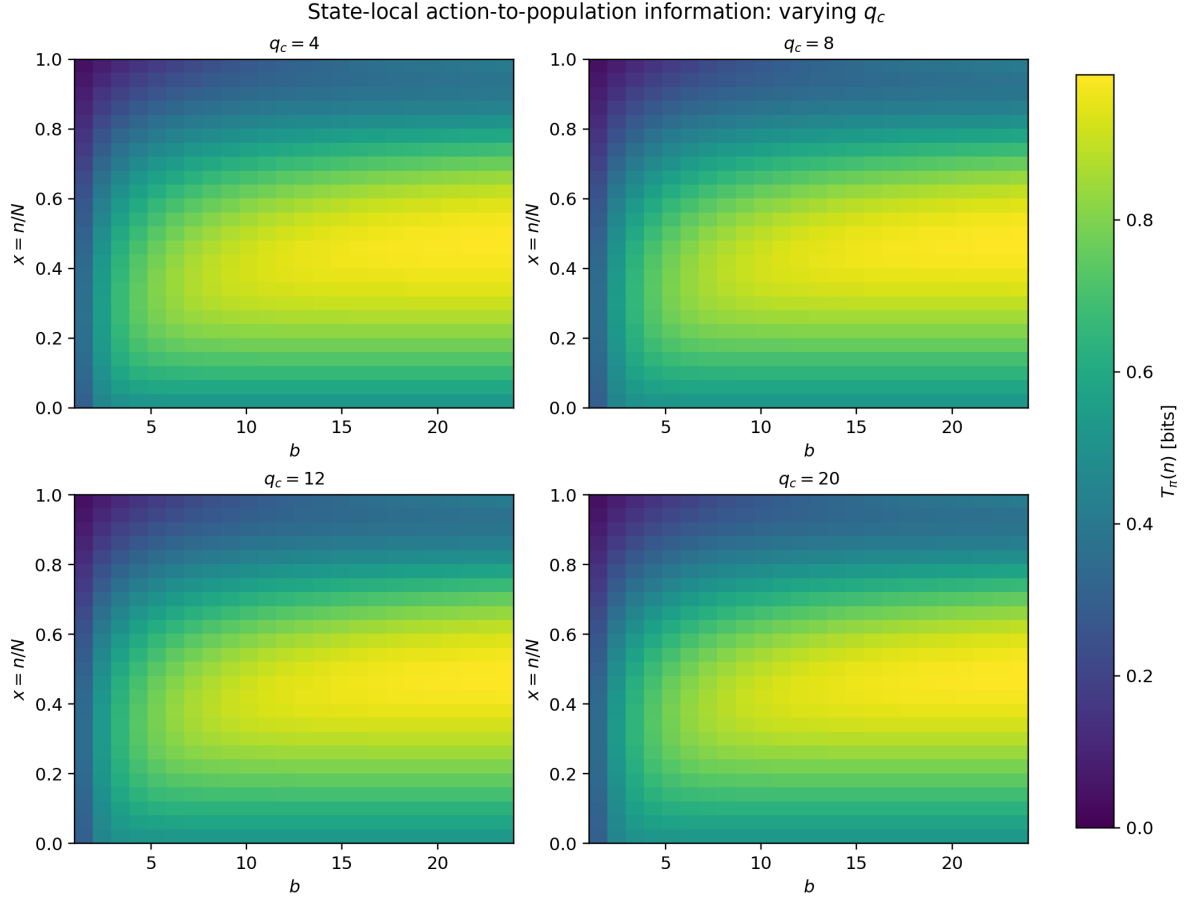


Figure 4: Exact state-local action-to-population information while varying  $q_c$ . The sensor sample size changes how sharply the controller resolves the current target fraction, thereby changing the state dependence of  $a_n$  and the ridge structure of  $T_\pi(n)$ .

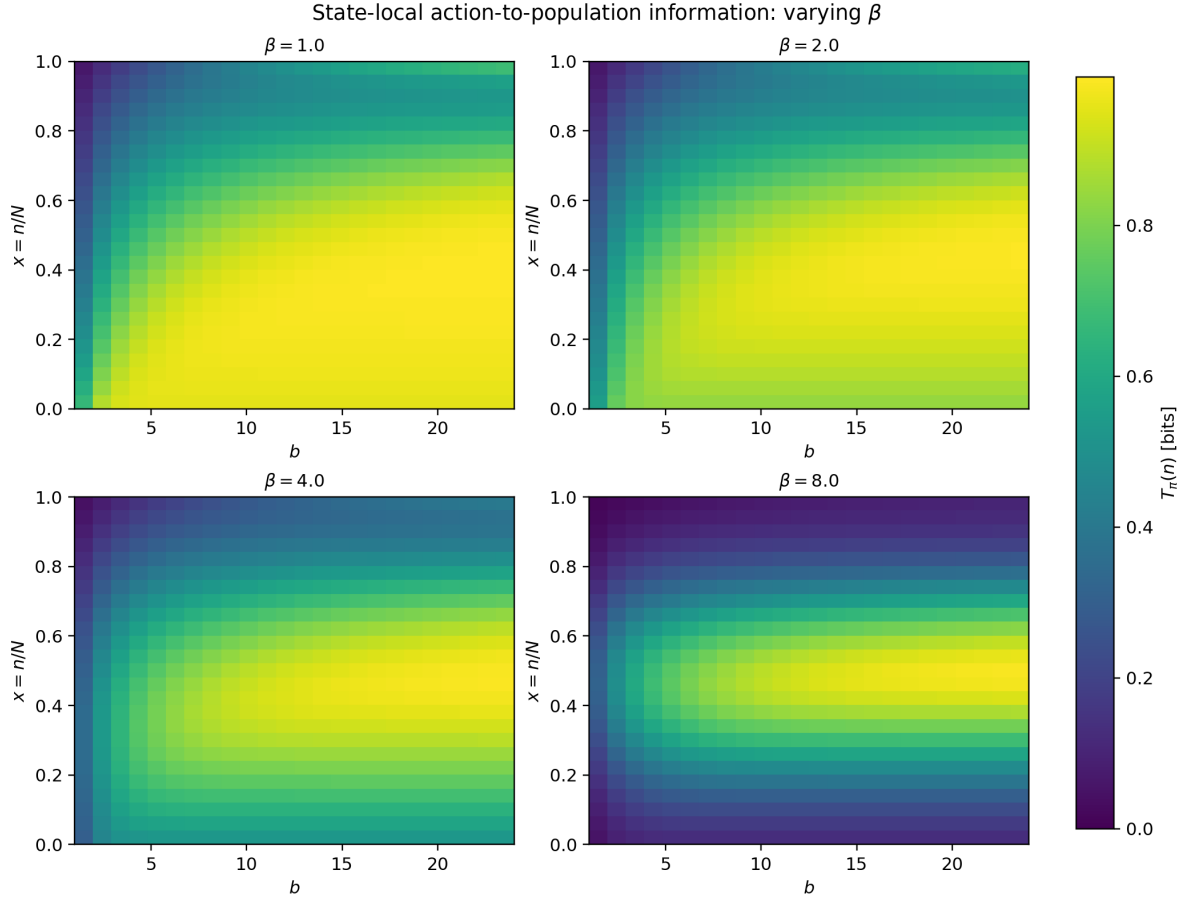


Figure 5: Exact state-local action-to-population information while varying  $\beta$ . Larger  $\beta$  makes the policy more switch-like: this can sharpen the informative band around the switching region while suppressing information where the action saturates.

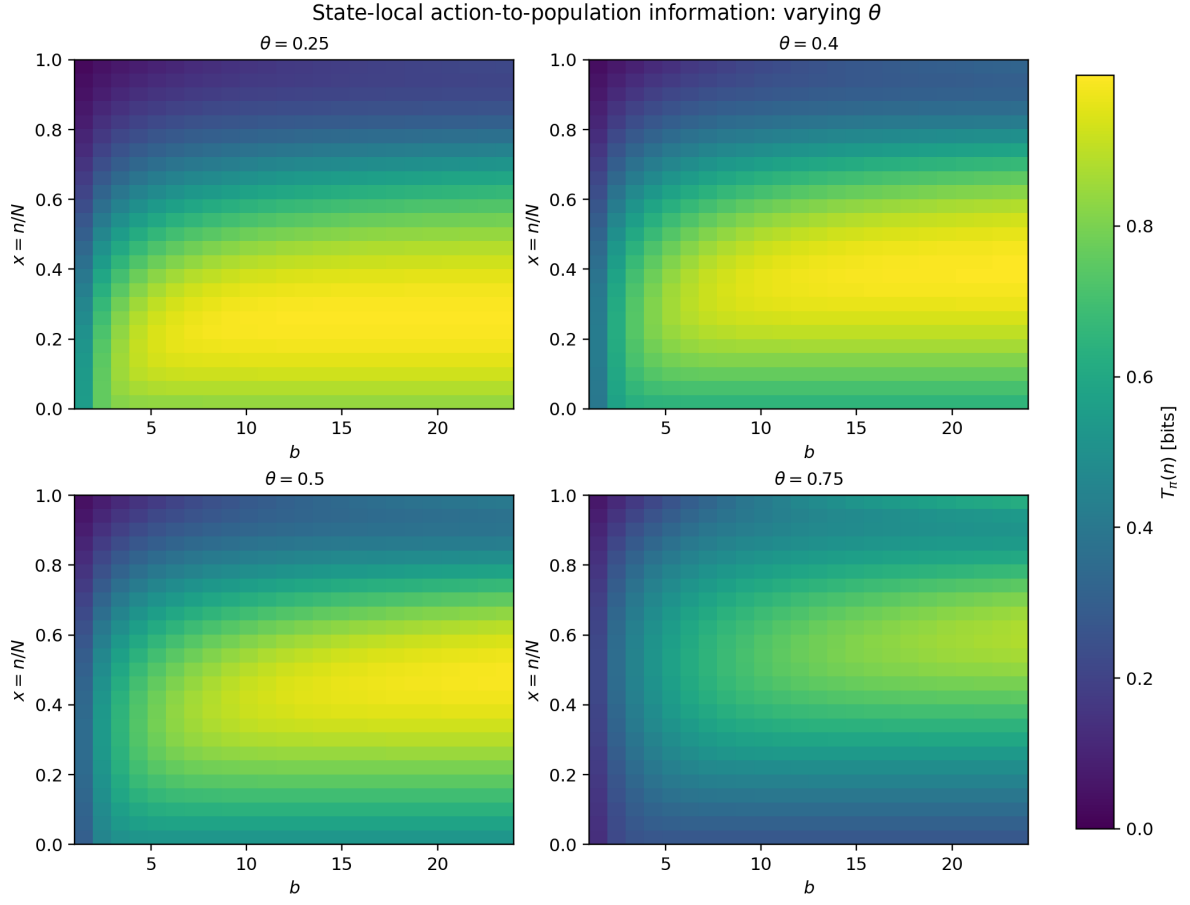


Figure 6: Exact state-local action-to-population information while varying  $\theta$ . The threshold shifts the state-space region where the controller is undecided and therefore shifts the high-information ridge.

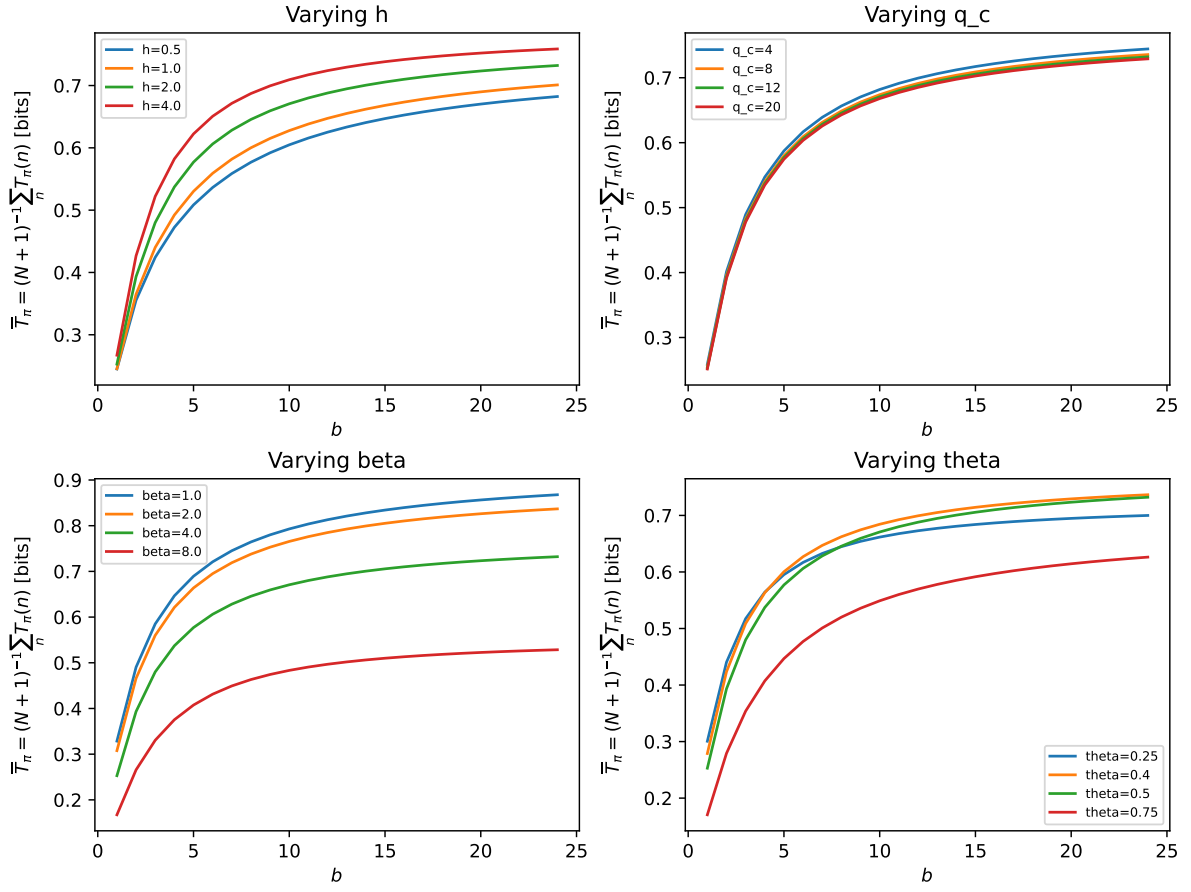


Figure 7: Arithmetic state-average  $\bar{T}_\pi = (N+1)^{-1} \sum_n T_\pi(n)$  for the four parameter sweeps. This compact view complements the heatmaps; it does not assume a population ensemble.

Define the combinatorial macrostate entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}. \quad (56)$$

Because

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1}, \quad (57)$$

we obtain

$$\boxed{\ln \frac{K_{h,\gamma}(n+1 | n)}{K_{h,\gamma}(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h.} \quad (58)$$

The compliance factor cancels exactly. Directionality and kinetics therefore remain distinct even at the path level.

Define the positive weight

$$w_h(n) = \binom{N}{n} e^{hn}. \quad (59)$$

The neighboring-state relation is equivalent to

$$\boxed{w_h(n)K_{h,\gamma}(m | n) = w_h(m)K_{h,\gamma}(n | m)} \quad (60)$$

for every supported transition. For diagonal transitions the equality is immediate; for  $m = n \pm 1$  it follows from the ratio just derived.

This weighted reversibility relation is preserved by matrix multiplication. Therefore, with  $Q_1 = K_{h,\gamma}^b$ ,

$$w_h(n)Q_1(m | n) = w_h(m)Q_1(n | m), \quad (61)$$

and hence

$$\boxed{\ln \frac{Q_1(m | n)}{Q_1(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h(m - n).} \quad (62)$$

This endpoint identity is what allows the finite-time path derivation to be written without resolving every controlled microscopic jump.

## 9 Forward and reverse finite-time feedback paths

For feedback cycle  $k$ , the forward path measure over the variables  $(n, y, u, m)$  is

$$\boxed{P_F^{(k)}(n, y, u, m) = p_k(n)S(y | n)\pi(u | y)Q_u(m | n).} \quad (63)$$

The forward final population distribution is obtained by marginalization:

$$\boxed{p_{k+1}(m) = \sum_{n,y,u} P_F^{(k)}(n, y, u, m).} \quad (64)$$

Equivalently, after averaging over the sensor and policy,

$$p_{k+1}(m) = \sum_n p_k(n) [(1 - a_n)\delta_{mn} + a_n Q_1(m | n)]. \quad (65)$$

The sensor marginal at the beginning of the cycle is

$$p_{Y,k}(y) = \sum_n p_k(n)S(y | n). \quad (66)$$

Use the reverse reference experiment

$$\boxed{P_R^{(k)}(m, y, u, n) = p_{k+1}(m)p_{Y,k}(y)\pi(u | y)Q_u(n | m).} \quad (67)$$

This construction has a direct operational reading. The reverse reference starts from the *forward final ensemble*  $p_{k+1}$ , samples a stored sensor label  $y$  independently from its forward marginal  $p_{Y,k}$ , applies the same policy law  $\pi(u | y)$ , and uses the reverse of the same controlled transition kernel. Because the same policy factor appears in the forward and reverse path weights, it cancels from their ratio.

Crucially, the reverse experiment does not assume stationarity. It starts from  $p_{k+1}$  whether or not that distribution equals  $p_k$ .

## 10 Finite-time path irreversibility and second-law identity

Define the stochastic path log-ratio

$$\boxed{\Sigma_k^* = \ln \frac{P_F^{(k)}(n, y, u, m)}{P_R^{(k)}(m, y, u, n)}.} \quad (68)$$

Substituting the path measures gives

$$\Sigma_k^* = \ln \frac{p_k(n)}{p_{k+1}(m)} + \ln \frac{S(y | n)}{p_{Y,k}(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}. \quad (69)$$

The factor  $\pi(u | y)$  cancels exactly.

Define the pointwise sensing information

$$i_k(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (70)$$

and the stochastic system entropy

$$s_{\text{sys}}(n; p_k) = -\ln p_k(n) + S_{\text{mix}}(n). \quad (71)$$

The controlled current increment is

$$j_c = u(m - n). \quad (72)$$

For  $u = 0$ ,  $Q_0 = I$ , so  $m = n$  and  $j_c = 0$ . For  $u = 1$ , the transition-ratio identity applies. Therefore

$$\ln \frac{p_k(n)}{p_{k+1}(m)} + S_{\text{mix}}(m) - S_{\text{mix}}(n) = s_{\text{sys}}(m; p_{k+1}) - s_{\text{sys}}(n; p_k), \quad (73)$$

and we obtain the central pathwise identity

$$\boxed{\Sigma_k^* = \Delta s_{\text{sys}} + h j_c + i_k(n; y).} \quad (74)$$

Average over the forward path ensemble. By construction,

$$\Sigma_k \equiv \mathbb{E}_F[\Sigma_k^*] = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \quad (75)$$

The three average terms are

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k], \quad (76)$$

$$J_{c,k} = \mathbb{E}_F[j_c], \quad (77)$$

$$I(n_k; Y_k) = \mathbb{E}_F[i_k(n; Y)]. \quad (78)$$

The system-entropy functional is

$$S_{\text{sys}}[p] = - \sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}. \quad (79)$$

The first term is the Shannon entropy of the coarse population state and the second averages the internal combinatorial multiplicity of each macrostate.

The sensing information has the finite-sum form

$$I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}. \quad (80)$$

Combining the averages yields the main finite-time stochastic-thermodynamic statement:

$$\Delta S_{\text{sys},k} + hJ_{c,k} + I(n_k; Y_k) = \Sigma_k = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \quad (81)$$

This is a statement about finite-time path irreversibility of the coarse-grained feedback process. It is not a work balance. No equilibrium state, steady state, heat bath, or energetic load is required for the identity.

The finite-time entropy change is essential. If the controller shifts the population ensemble during a cycle, then  $p_{k+1} \neq p_k$  and  $\Delta S_{\text{sys},k}$  need not vanish. Dropping this term would silently impose a condition that the derivation does not need.

Because  $\Sigma_k$  is a KL divergence, the identity immediately gives the finite-time second-law-like bound

$$\Delta S_{\text{sys},k} + hJ_{c,k} + I(n_k; Y_k) \geq 0, \quad (82)$$

or, equivalently,

$$hJ_{c,k} + I(n_k; Y_k) \geq -\Delta S_{\text{sys},k}. \quad (83)$$

This is the fundamental thermodynamic inequality. The bounded efficiency below is a consequence of the same accounting after the transient storage term is separated from the feedback expenditure.

## 11 Finite-time thermodynamic control efficiency

The path identity contains two conceptually different kinds of terms. The system-entropy change  $\Delta S_{\text{sys},k}$  is a finite-time boundary or storage contribution: it records that the population ensemble itself may change during the cycle. To isolate the part of the accounting associated with the feedback operation rather than transient storage, define the non-storage finite-time control expenditure

$$C_{\text{th},k} \equiv \Sigma_k - \Delta S_{\text{sys},k}. \quad (84)$$

Using the exact path identity,

$$C_{\text{th},k} = hJ_{c,k} + I(n_k; Y_k). \quad (85)$$

The two terms have distinct operational meanings. The affinity-weighted current  $hJ_{c,k}$  is the directed population motion produced by actuation, whereas  $I(n_k; Y_k)$  is the information acquired by sensing before the action is selected. No energetic work interpretation is required.

For target-directed operation,  $hJ_{c,k} \geq 0$ . Whenever  $C_{\text{th},k} > 0$ , the fraction of the non-storage control expenditure expressed as directed current is therefore

$$\eta_{\text{th},k} \equiv \frac{hJ_{c,k}}{C_{\text{th},k}} = \frac{hJ_{c,k}}{\Sigma_k - \Delta S_{\text{sys},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I(n_k; Y_k)} \leq 1. \quad (86)$$



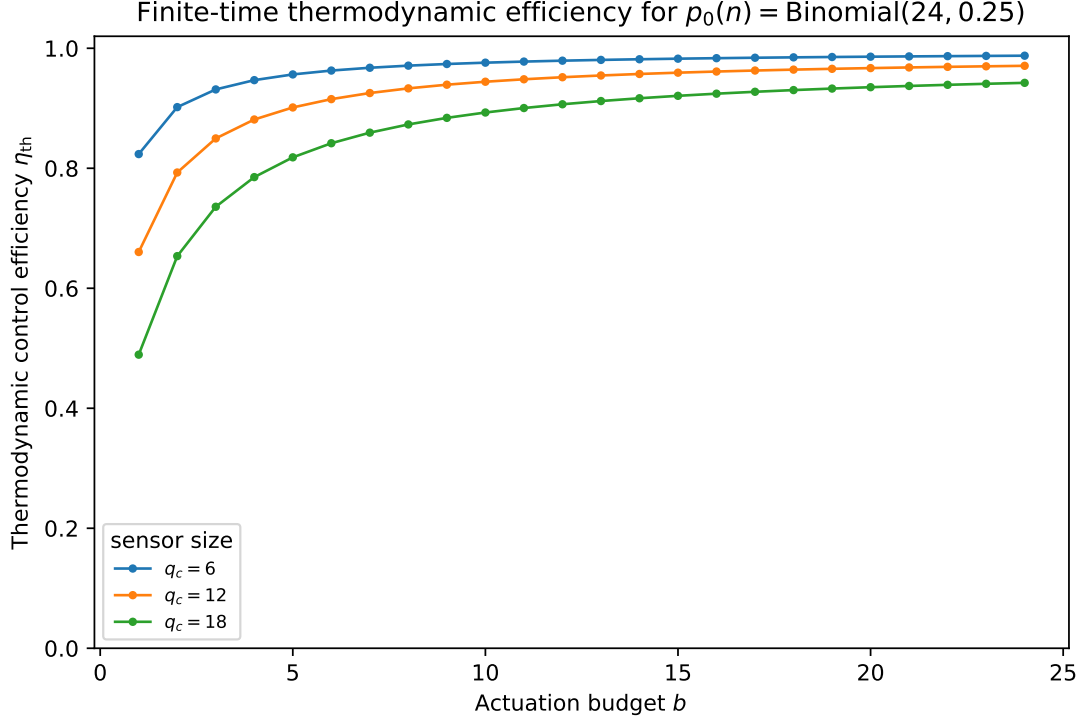


Figure 8: Exact finite-time thermodynamic control efficiency  $\eta_{th} = hJ_c/[hJ_c + I(n; Y)]$  for  $p_0(n) = \text{Binomial}(24, 0.25)$ ,  $h = 2$ ,  $\gamma = 0.75$ ,  $\beta = 4$ , and  $\theta = 1/2$ . Each curve varies the actuation budget  $b$  at fixed sensing budget  $q_c$ . Increasing actuation raises the affinity-weighted current, while increasing sensing lowers the efficiency when the additional sensing information is not accompanied by a proportional current increase.

Since  $I(n_k; Y_k) \geq 0$ , the target-directed regime obeys

$$0 \leq \eta_{th,k} \leq 1 \quad (hJ_{c,k} \geq 0, C_{th,k} > 0). \quad (87)$$

If  $C_{th,k} = 0$ , the ratio is undefined rather than artificially set to zero. If  $hJ_{c,k} < 0$ , the signed ratio can still be reported diagnostically, but it no longer has the interpretation of a bounded target-directed efficiency.

At stationarity the boundary term vanishes,  $\Delta S_{sys,k} = 0$ , and therefore

$$\eta_{th} = \frac{hJ_c}{\Sigma}. \quad (88)$$

The finite-time expression is not obtained by assuming stationarity; it removes the transient storage term explicitly before forming the ratio.

The resource trade-off is transparent. Holding the sensor channel fixed, increasing actuation can raise  $hJ_c$  and drive  $\eta_{th}$  upward. Conversely, increasing  $q_c$  can increase the sensing information without a proportional increase in directed current, which lowers the efficiency. Thus more sensing is not intrinsically undesirable; it becomes thermodynamic overhead only when the additional acquired information is not converted into additional target-directed motion.

Figure 8 shows this exact prediction for the sensing budgets used in the main resource grid. It uses the same illustrative finite-time ensemble  $p_0(n) = \text{Binomial}(24, 0.25)$ , with  $h = 2$ ,  $\gamma = 0.75$ ,  $\beta = 4$ , and  $\theta = 1/2$ . At  $b = 18$ , the resulting efficiencies are approximately 0.985, 0.964, and 0.930 for  $q_c = 6, 12, 18$ , respectively.

### 11.1 Current–irreversibility view as a complementary diagnostic

The bounded efficiency does not replace the direct current–irreversibility geometry. A controller configuration can still be characterized by the pair

$$\boxed{(J_c, \Sigma)} \quad (89)$$

or, when the response at a particular state is the relevant observable, by

$$\boxed{(\chi, \Sigma)}. \quad (90)$$

A direct control-design formulation is

$$\boxed{\pi^* = \arg \min_{\pi} \Sigma(\pi) \quad \text{subject to} \quad J_c(\pi) \geq J_{\text{req}},} \quad (91)$$

or equivalently

$$\boxed{\pi^* = \arg \max_{\pi} J_c(\pi) \quad \text{subject to} \quad \Sigma(\pi) \leq \Sigma_{\text{max}}.} \quad (92)$$

These Pareto formulations answer a different design question from  $\eta_{\text{th}}$ : they retain the complete path irreversibility, including the transient system-entropy contribution, rather than normalizing out storage.

Figure 9 preserves this view for the same type of finite-time ensemble. It uses

$$p_0(n) = \text{Binomial}(24, 0.25), \quad (93)$$

with  $h = 2$ ,  $\gamma = 0.75$ ,  $\beta = 4$ , and  $\theta = 1/2$ . Each curve varies the actuation budget  $b = 1, \dots, 24$ , while the curves compare different sensor sizes  $q_c$ . The corresponding data file also records the direct KL calculation and the decomposition  $\Delta S_{\text{sys}} + hJ_c + I(n; Y)$ ; the two agree numerically to floating-point precision.

## 12 Direct finite-sum computation recipe

For one cycle with pre-sensing ensemble  $p_k(n)$ , every theoretical quantity above is computable by finite sums and a finite matrix power:

1. Compute the sensor kernel  $S(y | n)$  and policy  $\pi(1 | y)$ .
2. Compute

$$a_n = \sum_y S(y | n) \pi(1 | y), \quad p_{Y,k}(y) = \sum_n p_k(n) S(y | n).$$

3. Set  $p_h = \sigma(h)$  and

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b.$$

4. Build  $K_{h,\gamma}$  and  $Q_1 = K_{h,\gamma}^b$ , with  $Q_0 = I$ .
5. Compute the mean controlled current

$$J_{c,k} = N \Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[ p_h - \frac{n}{N} \right].$$

6. Compute the sensing information

$$I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}.$$

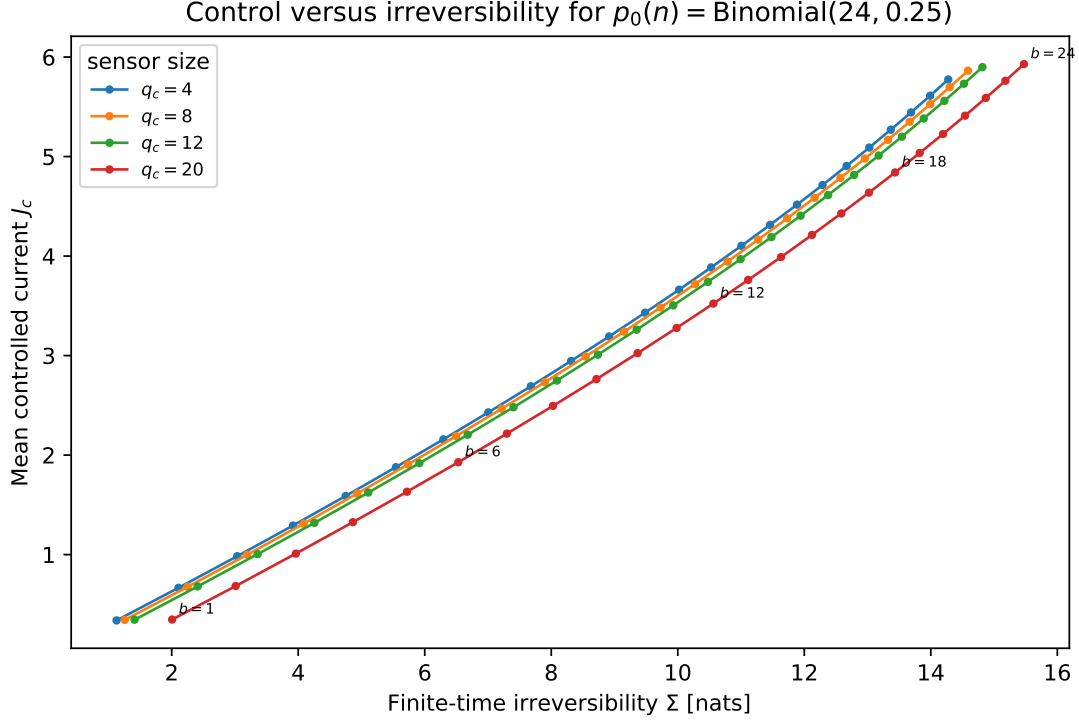


Figure 9: Mean controlled current versus finite-time irreversibility for an illustrative initial ensemble  $p_0(n) = \text{Binomial}(24, 0.25)$ . Each curve varies  $b$  from 1 to 24 at fixed  $h = 2$ ,  $\gamma = 0.75$ ,  $\beta = 4$ , and  $\theta = 1/2$ ; labels mark selected  $b$  values. Different sensing resources can yield nearly the same controlled current while requiring different total path irreversibility because the sensing-information contribution changes.

7. Propagate the ensemble exactly:

$$p_{k+1}(m) = \sum_n p_k(n) [(1 - a_n)\delta_{mn} + a_n Q_1(m | n)].$$

8. Compute

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k].$$

9. Evaluate the path irreversibility

$$\Sigma_k = \Delta S_{\text{sys},k} + hJ_{c,k} + I(n_k; Y_k).$$

As a numerical check, compute the path measures directly and verify

$$\Sigma_k = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}).$$

10. Separate the non-storage control expenditure,

$$C_{\text{th},k} = \Sigma_k - \Delta S_{\text{sys},k} = hJ_{c,k} + I(n_k; Y_k).$$

For  $hJ_{c,k} \geq 0$  and  $C_{\text{th},k} > 0$ , compute

$$\eta_{\text{th},k} = \frac{hJ_{c,k}}{C_{\text{th},k}} = \frac{hJ_{c,k}}{hJ_{c,k} + I(n_k; Y_k)}.$$

There is no Monte Carlo approximation in this reference calculation. Monte Carlo or direct counting enters only when the same objects are estimated from LLM trajectories rather than generated by the finite-state reference kernel.

## 12.1 How to estimate the terms from repeated trajectories

At a fixed cycle  $k$ ,  $p_k(n)$  and  $p_{k+1}(m)$  can be estimated as empirical histograms across repeated episodes. The sensor kernel is known exactly under the sampling protocol, while the logged policy probabilities or action frequencies estimate  $a_n$ . The controlled current can be estimated directly as the sample mean of  $U_k(n_{k+1} - n_k)$ , or from the calibrated reference kernel. Sensing information is a contingency-table mutual information between  $n_k$  and  $Y_k$ ; if an empirical analysis reports it in bits, multiply by  $\ln 2$  before inserting it into the natural-log thermodynamic identity. The state-local  $T_\pi(n)$  is a separate conditional mutual information and can remain in bits for comparison with the empirical action channel.

Estimating the full path KL directly from sparse LLM data is more demanding because it requires support for both forward and reverse coarse-grained path probabilities. The equality above therefore provides a useful model-based decomposition, while empirical support and overlap diagnostics remain necessary when conditional channels are estimated by direct counting. Once  $h$ ,  $J_c$ , and the sensing information are estimated in consistent natural-log units, the non-storage expenditure and  $\eta_{\text{th}}$  can be evaluated without estimating the full path KL separately; the direct KL remains the strongest numerical validation of the reference model.

## 13 Role of kinetic compliance

The kinetic parameter  $\gamma$  controls the speed and amplitude of the response, not its directional affinity. At the mean-response level it appears through

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (94)$$

At the path level it changes  $Q_1 = K_{h,\gamma}^b$  and therefore changes the final ensemble  $p_{k+1}$  and the path KL divergence. But it cancels from the forward/reverse microscopic ratio because the same factor multiplies both directions. There is therefore no additional  $\ln \gamma$  contribution to the directional affinity.

At  $\gamma = 0$ ,  $K_{h,0} = I$  and  $J_c = 0$ : the controller has no kinetic actuation. For  $0 < \gamma \leq 1$ , the full finite-time derivation applies. In the large- $h$  limit, directionality becomes nearly one-sided, but actuation can still remain incomplete when  $\gamma < 1$ .

## 14 Operational calibration from microscopic LLM transitions

Controlled microscopic transitions provide a direct operational calibration of both model parameters. Define

$$p_+ = P(\text{non-}Z \rightarrow Z \mid \text{controlled}), \quad (95)$$

$$p_- = P(Z \rightarrow \text{non-}Z \mid \text{controlled}). \quad (96)$$

Under the finite-compliance kernel,

$$p_+ = \gamma \sigma(h), \quad p_- = \gamma \sigma(-h). \quad (97)$$

Adding the two probabilities gives

$$p_+ + p_- = \gamma[\sigma(h) + \sigma(-h)] = \gamma, \quad (98)$$

while their ratio gives

$$\frac{p_+}{p_-} = \frac{\sigma(h)}{\sigma(-h)} = e^h. \quad (99)$$

Hence

$$\boxed{\gamma_{\text{eff}} = p_+ + p_-, \quad h_{\text{eff}} = \ln \frac{p_+}{p_-}.} \quad (100)$$

Using the existing controlled-transition counts,

$$p_+ = \frac{208}{572} \approx 0.364, \quad p_- = \frac{4}{508} \approx 0.0079, \quad (101)$$

we recover

$$\boxed{\gamma_{\text{eff}} \approx 0.372, \quad h_{\text{eff}} \approx 3.83.} \quad (102)$$

This is a complete operational calibration of the two parameters in the simplified controlled kernel: one number describes kinetic compliance and the other describes directional asymmetry.

## 15 Scope and modeling limitations

The theory should be read as a finite-state coarse-grained reference model for the controlled population coordinate  $n_k$ . It is not a claim of a complete thermodynamics of all social interactions among language-model agents.

The derivation is explicitly finite-time. It does not assume equilibrium or stationarity. The changing population ensemble is represented by

$$p_k \longrightarrow p_{k+1}, \quad (103)$$

and contributes

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k]. \quad (104)$$

This is why the reverse reference begins from the forward final ensemble rather than from the initial ensemble.

Finally,  $n_k$  is a coarse-grained state variable. In the LLM system, future transitions may depend on hidden epistemic state, private fact sets, dialogue history, persistent reasons, or other variables not encoded in the target count. The empirical reduced process in  $n_k$  can therefore be imperfectly Markovian. This is a modeling limitation to diagnose, not an assumption to hide. The finite-state kernel should consequently be used as a matched reference model whose deviations from the LLM trajectories are scientifically informative.

**What each observable means.** The sensing term  $I(n_k; Y_k)$  asks how much the controller learns about the current population. The susceptibility  $\chi_{h,\gamma}(x)$  asks for the signed mean response at a given state. The local information  $T_\pi(n)$  asks how much the next population state reveals about the controller action, while  $\eta_{\text{IR}}(n)$  asks how coherently that action-dependent distributional change appears as mean target motion. The current  $J_{c,k}$  aggregates the target-directed response over the occupied population states. The path irreversibility  $\Sigma_k$  quantifies the finite-time statistical irreversibility of the complete coarse-grained feedback cycle relative to the specified reverse reference. Separating its transient storage term gives  $C_{\text{th},k}$  and the bounded target-directed efficiency  $\eta_{\text{th},k}$ . These observables therefore retain distinct roles: sensing, action visibility, coherent response, directed current, path irreversibility, and finite-time thermodynamic conversion are not collapsed into one undifferentiated number.

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